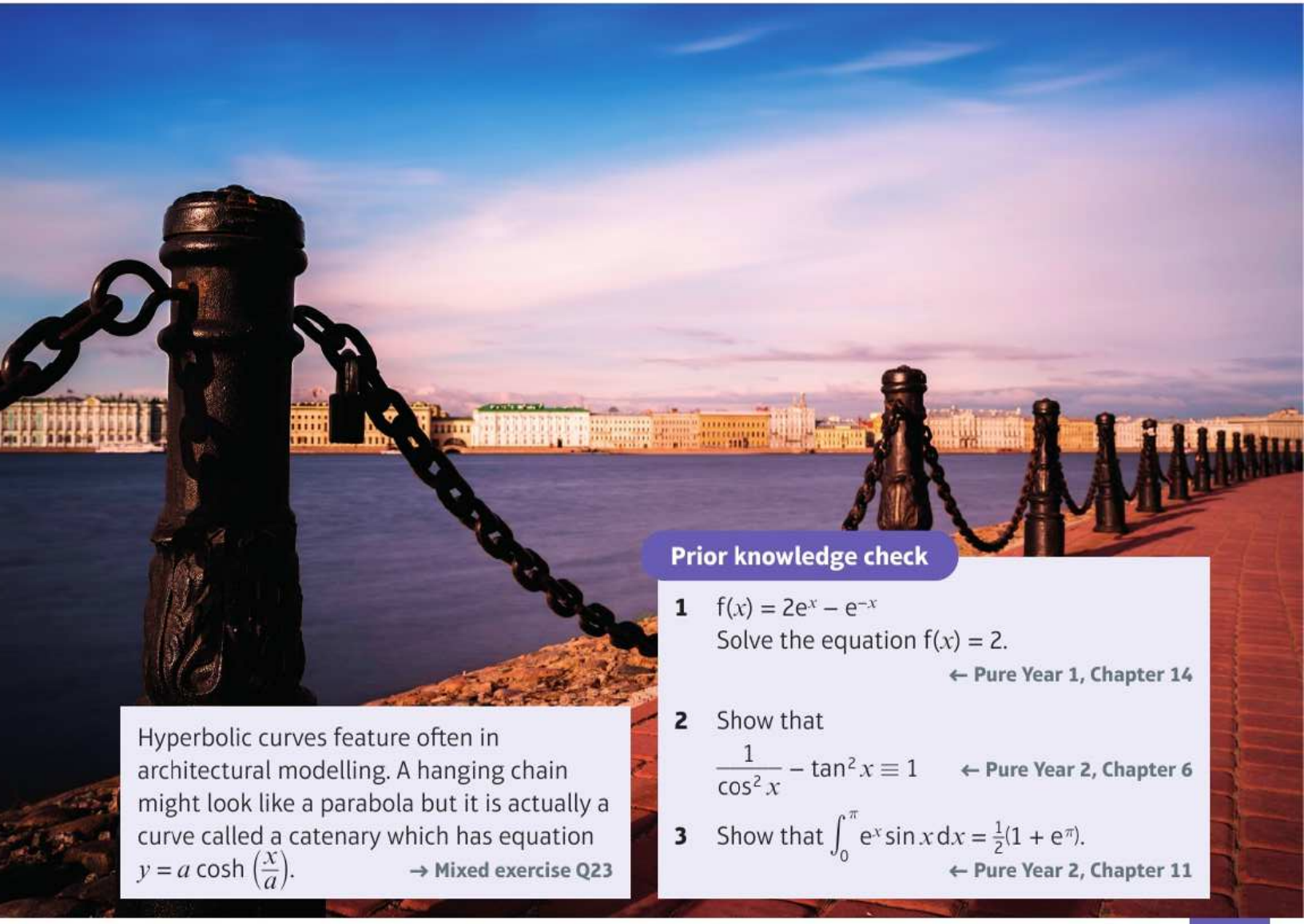
Hyperbolic functions

6

Objectives

After completing this chapter you should be able to:

- Understand the definitions of hyperbolic functions → pages 120–123
- Sketch the graphs of hyperbolic functions → pages 121–123
- Understand and use the inverse hyperbolic functions → pages 123–125
- Prove identities and solve equations using hyperbolic functions → pages 125–129
- Differentiate and integrate hyperbolic functions → pages 130–142



Hyperbolic curves feature often in architectural modelling. A hanging chain might look like a parabola but it is actually a curve called a catenary which has equation $y = a\cosh\left(\frac{x}{a}\right)$. → Mixed exercise Q23

Prior knowledge check

1 $f(x) = 2e^x - e^{-x}$
Solve the equation $f(x) = 2$.
← Pure Year 1, Chapter 14

2 Show that
$\frac{1}{\cos^2 x} - \tan^2 x \equiv 1$ ← Pure Year 2, Chapter 6

3 Show that $\int_0^{\pi} e^x \sin x \, dx = \frac{1}{2}(1 + e^{\pi})$.
← Pure Year 2, Chapter 11

6.1 Introduction to hyperbolic functions

Hyperbolic functions have several properties in common with trigonometric functions, but they are defined in terms of exponential functions.

- **Hyperbolic sine (or sinh) is defined as** $\sinh x \equiv \dfrac{e^x - e^{-x}}{2}$
- **Hyperbolic cosine (or cosh) is defined as** $\cosh x \equiv \dfrac{e^x + e^{-x}}{2}$
- **Hyperbolic tangent (or tanh) is defined as** $\tanh x \equiv \dfrac{\sinh x}{\cosh x}$

You can use the definitions of $\sinh x$ and $\cosh x$ to write $\tanh x$ in exponential form.

$$\tanh x \equiv \frac{\sinh x}{\cosh x} \equiv \frac{e^x - e^{-x}}{2} \times \frac{2}{e^x + e^{-x}} \equiv \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Multiplying the numerator and denominator of the final expression through by e^x gives:

- $\tanh x \equiv \dfrac{e^{2x} - 1}{e^{2x} + 1}$

There are also hyperbolic functions corresponding to the reciprocal trigonometric functions:

$$\operatorname{cosech} x \equiv \frac{1}{\sinh x} \equiv \frac{2}{e^x - e^{-x}}$$

$$\operatorname{sech} x \equiv \frac{1}{\cosh x} \equiv \frac{2}{e^x + e^{-x}}$$

$$\coth x \equiv \frac{1}{\tanh x} \equiv \frac{e^{2x} + 1}{e^{2x} - 1}$$

Note You won't need to use these functions in your exam, but they are useful to know, and if you are confident with them you can use them to simplify your working.

Example 1

Find, to 2 decimal places, the values of:

a $\sinh 3$ **b** $\cosh 1$ **c** $\tanh 0.8$

a $\sinh 3 = \dfrac{e^3 - e^{-3}}{2} = 10.02$ (2 d.p.)

b $\cosh 1 = \dfrac{e^1 + e^{-1}}{2} = 1.54$ (2 d.p.)

c $\tanh 0.8 = \dfrac{e^{1.6} - 1}{e^{1.6} + 1} = 0.66$ (2 d.p.)

Example 2

Find the exact value of $\tanh(\ln 4)$.

$$\tanh(\ln 4) = \frac{e^{2\ln 4} - 1}{e^{2\ln 4} + 1} = \frac{e^{\ln 4^2} - 1}{e^{\ln 4^2} + 1} = \frac{e^{\ln 16} - 1}{e^{\ln 16} + 1}$$

$$= \frac{16 - 1}{16 + 1} = \frac{15}{17}$$

Use $e^{\ln k} = k$.

Example 3

Use the definition of $\sinh x$ to find, to 2 decimal places, the value of x for which $\sinh x = 5$.

$\frac{e^x - e^{-x}}{2} = 5 \Rightarrow e^x - e^{-x} = 10$

$e^{2x} - 1 = 10e^x$ — Multiply both sides by e^x.

$e^{2x} - 10e^x - 1 = 0$ — The substitution $u = e^x$ turns this into the quadratic equation $u^2 - 10u - 1 = 0$.

$e^x = 5 \pm \sqrt{26}$

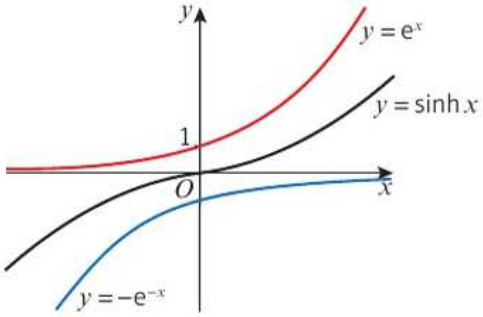
$\Rightarrow e^x = 5 + \sqrt{26}$ — e^x cannot be negative.

So $x = \ln(5 + \sqrt{26}) = 2.31$ (2 d.p.)

You can sketch the graphs of the hyperbolic functions by considering the graphs of $y = e^x$ and $y = e^{-x}$.

$$\sinh x = \frac{e^x - e^{-x}}{2} = \frac{e^x + (-e^{-x})}{2}$$

so the graph of $y = \sinh x$ is the 'average' of the graphs of $y = e^x$ and $y = -e^{-x}$.

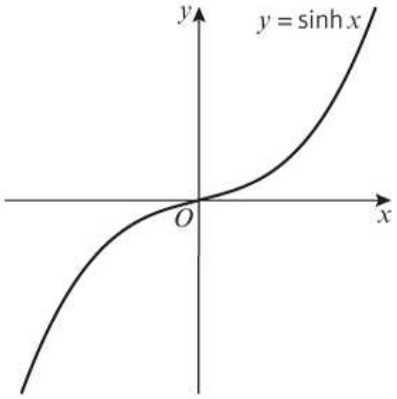


For the graph of $y = \sinh x$,

- when x is large and positive, e^{-x} is small, so $\sinh x \approx \frac{1}{2}e^x$
- when x is large and negative, e^x is small, so $\sinh x \approx -\frac{1}{2}e^{-x}$

- **For any value a, $\sinh(-a) = -\sinh a$**

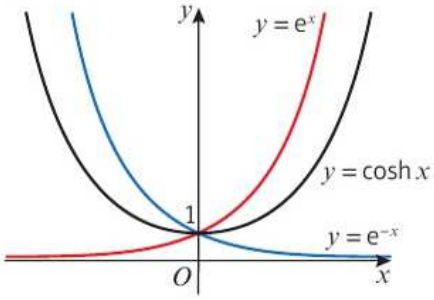
Notation $f(x) = \sinh x$ is an **odd** function since $f(-x) = -f(x)$.



Consider the graphs of $y = e^x$ and $y = e^{-x}$.

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

so the graph of $y = \cosh x$ is the 'average' of the graphs of $y = e^x$ and $y = e^{-x}$.

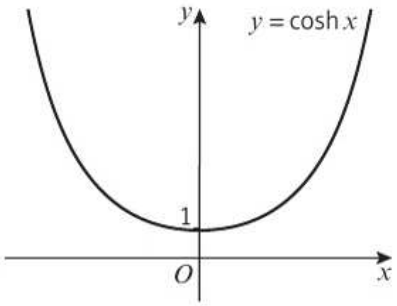


For the graph of $y = \cosh x$,

- when x is large and positive, e^{-x} is small, so $\cosh x \approx \frac{1}{2}e^x$
- when x is large and negative, e^x is small, so $\cosh x \approx \frac{1}{2}e^{-x}$

- **For any value a, $\cosh(-a) = \cosh a$**

Notation $f(x) = \cosh x$ is an **even** function because $f(-x) = f(x)$.



Example 4

Sketch the graph of $y = \tanh x$.

$\tanh x = \dfrac{\sinh x}{\cosh x}$

When $x = 0$, $\tanh x = \frac{0}{1} = 0$

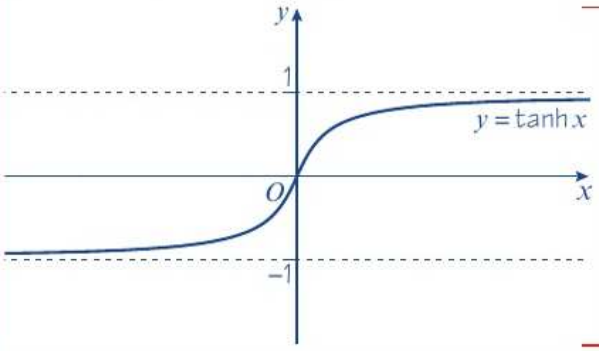
When x is large and positive, $\sinh x \approx \frac{1}{2}e^x$ and $\cosh x \approx \frac{1}{2}e^x$, so $\tanh x \approx 1$.

When x is large and negative, $\sinh x \approx -\frac{1}{2}e^{-x}$ and $\cosh x \approx \frac{1}{2}e^{-x}$, so $\tanh x \approx -1$.

As $x \to \infty$, $\tanh x \to 1$ and as $x \to -\infty$, $\tanh x \to -1$

For $f(x) = \tanh x$, $x \in \mathbb{R}$, the range of f is $-1 < f(x) < 1$

$y = -1$ and $y = 1$ are asymptotes to the curve.



Online Explore graphs of hyberbolic functions using GeoGebra.



Consider the graphs of $y = \sinh x$ and $y = \cosh x$ to work out the behaviour of $y = \tanh x$ as $x \to \infty$ and $x \to -\infty$.

You should always include any asymptotes on a sketch graph.

Exercise 6A

1 Find, correct to 2 decimal places:

a $\sinh 4$ **b** $\cosh \frac{1}{2}$ **c** $\tanh(-2)$

2 Write, in terms of e:

a $\sinh 1$ **b** $\cosh 4$ **c** $\tanh 0.5$

3 Find the exact values of:

a $\sinh(\ln 2)$ **b** $\cosh(\ln 3)$ **c** $\tanh(\ln 2)$

In questions **4** to **6**, use the definitions of the hyperbolic functions (in terms of exponentials) to find each answer, then check your answers using an inverse hyperbolic function on your calculator.

4 Find, to 2 decimal places, the values of x for which $\cosh x = 2$.

5 Find, to 2 decimal places, the values of x for which $\sinh x = 1$.

6 Find, to 2 decimal places, the values of x for which $\tanh x = -\frac{1}{2}$

7 On the same diagram, sketch the graphs of $y = \cosh 2x$ and $y = 2\cosh x$.

8 Find the range of each hyperbolic function.

a $f(x) = \sinh x, x \in \mathbb{R}$

b $f(x) = \cosh x, x \in \mathbb{R}$

c $f(x) = \tanh x, x \in \mathbb{R}$

E **9 a** Sketch the graph of $y = 3\tanh x + 2$. **(3 marks)**

b Write down the equations of the asymptotes to this curve. **(2 marks)**

Challenge Sketch the graphs of:

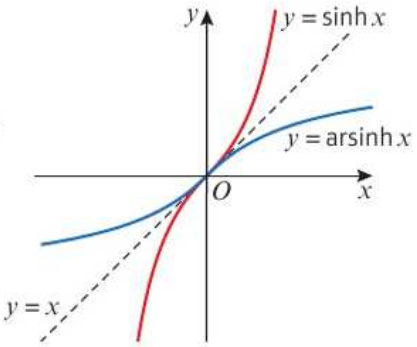
a $y = \text{sech}\, x$ **b** $y = \text{cosech}\, x$ **c** $y = \coth x$

6.2 Inverse hyperbolic functions

You can define and use the inverses of the hyperbolic functions.

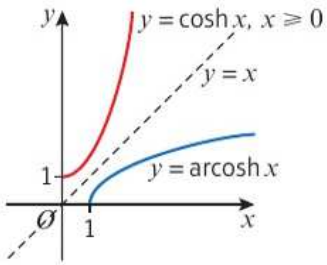
If $f(x) = \sinh x$, the inverse function f^{-1} is called arsinh x.

The graph of $y = \text{arsinh}\, x$ is the reflection of the graph of $y = \sinh x$ in the line $y = x$.



The inverse of a function is defined only if the function is one-to-one, so for $\cosh x$ the domain must be restricted in order to define an inverse.

For $f(x) = \cosh x, x \geqslant 0$, $f^{-1}(x) = \text{arcosh}\, x, x \geqslant 1$



- **The following table shows the inverse hyperbolic functions, with domains restricted where necessary.**

Hyperbolic function	Inverse hyperbolic function
$y = \sinh x$	$y = \text{arsinh}\, x$
$y = \cosh x, x \geqslant 0$	$y = \text{arcosh}\, x, x \geqslant 1$
$y = \tanh x$	$y = \text{artanh}\, x, \lvert x\rvert < 1$

Notation arsinh, arcosh and artanh are sometimes written as $\sinh^{-1}$, $\cosh^{-1}$ and $\tanh^{-1}$.

You can express the inverse hyperbolic functions in terms of natural logarithms.

Example 5

Show that $\operatorname{arsinh} x = \ln\left(x + \sqrt{x^2 + 1}\right)$.

Let $y = \operatorname{arsinh} x$

$x = \sinh y$

$x = \frac{e^y - e^{-y}}{2}$ — Use the definition of sinh.

$e^y - e^{-y} = 2x$

$e^{2y} - 1 = 2xe^y$ — Multiply by e^y.

$e^{2y} - 2xe^y - 1 = 0$

$(e^y - x)^2 - x^2 - 1 = 0$

$e^y = x \pm \sqrt{x^2 + 1}$

$e^y = x - \sqrt{x^2 + 1}$ can be ignored since $\sqrt{x^2 + 1} > x$, and would give a negative value of e^y, which is not possible.

So $e^y = x + \sqrt{x^2 + 1}$

$y = \ln\left(x + \sqrt{x^2 + 1}\right)$

$\Rightarrow \operatorname{arsinh} x = \ln\left(x + \sqrt{x^2 + 1}\right)$

Problem-solving

$e^{2y} - 2xe^y - 1 = 0$ is a quadratic in e^y.
You can write it as $(e^y)^2 - 2xe^y - 1 = 0$ and then complete the square.

Example 6

Show that $\operatorname{arcosh} x = \ln\left(x + \sqrt{x^2 - 1}\right)$, $x \geqslant 1$.

Let $y = \operatorname{arcosh} x$

$x = \cosh y$

$x = \frac{e^y + e^{-y}}{2}$ — Use the definition of cosh.

$e^y + e^{-y} = 2x$

$e^{2y} + 1 = 2xe^y$ — Multiply by e^y.

$e^{2y} - 2xe^y + 1 = 0$ — Form and solve a quadratic in e^y.

$(e^y - x)^2 - x^2 + 1 = 0$

So $e^y = x \pm \sqrt{x^2 - 1}$

$y = \ln\left(x \pm \sqrt{x^2 - 1}\right)$ — Note that both $x + \sqrt{x^2 - 1}$ and $x - \sqrt{x^2 - 1}$ are positive.

$\Rightarrow \operatorname{arcosh} x = \ln\left(x + \sqrt{x^2 - 1}\right)$ — $\operatorname{arcosh} x$ is always non-negative. For all values of $x > 1$, $x - \sqrt{x^2 - 1} < 1$ so the value of $\ln\left(x - \sqrt{x^2 - 1}\right)$ is negative.

You can use a similar method to express $\operatorname{artanh} x$ in terms of natural logarithms.

The following formulae are provided in the formula booklet and can be used directly unless you are asked to prove them .

- $\operatorname{arsinh} x = \ln\left(x + \sqrt{x^2 + 1}\right)$
- $\operatorname{arcosh} x = \ln\left(x + \sqrt{x^2 - 1}\right)$, $x \geqslant 1$
- $\operatorname{artanh} x = \frac{1}{2}\ln\left(\frac{1 + x}{1 - x}\right)$, $|x| < 1$

Example 7

Express as natural logarithms:

a $\operatorname{arsinh} 1$ **b** $\operatorname{arcosh} 2$ **c** $\operatorname{artanh} \frac{1}{3}$

a $\operatorname{arsinh} 1 = \ln\left(1 + \sqrt{1^2 + 1}\right) = \ln\left(1 + \sqrt{2}\right)$

b $\operatorname{arcosh} 2 = \ln\left(2 + \sqrt{2^2 - 1}\right) = \ln\left(2 + \sqrt{3}\right)$

c $\operatorname{artanh} \frac{1}{3} = \frac{1}{2}\ln\left(\frac{1 + \frac{1}{3}}{1 - \frac{1}{3}}\right) = \frac{1}{2}\ln 2 = \ln\sqrt{2}$

Use $a \ln x = \ln x^a$.

Exercise 6B

1 Sketch the graph of $y = \operatorname{artanh} x$, $|x| < 1$.

(P) **2** Sketch the graph of $y = (\operatorname{arsinh} x)^2$.

(E/P) **3** Prove that $\operatorname{artanh} x = \frac{1}{2}\ln\left(\frac{1 + x}{1 - x}\right)$, $|x| < 1$. **(5 marks)**

4 Express as natural logarithms:

a $\operatorname{arsinh} 2$ **b** $\operatorname{arcosh} 3$ **c** $\operatorname{artanh} \frac{1}{2}$

5 Express as natural logarithms:

a $\operatorname{arsinh} \sqrt{2}$ **b** $\operatorname{arcosh} \sqrt{5}$ **c** $\operatorname{artanh} 0.1$

6 Express as natural logarithms:

a $\operatorname{arsinh} (-3)$ **b** $\operatorname{arcosh} \frac{3}{2}$ **c** $\operatorname{artanh} \frac{1}{\sqrt{3}}$

(E/P) **7** Given that $\operatorname{artanh} x + \operatorname{artanh} y = \ln\sqrt{3}$, prove that $y = \dfrac{2x - 1}{x - 2}$. **(6 marks)**

6.3 Identities and equations

You can find and use identities for the hyperbolic functions that are similar to the trigonometric identities.

Example 8

Prove that $\cosh^2 A - \sinh^2 A \equiv 1$

$\text{LHS} \equiv \cosh^2 A - \sinh^2 A \equiv \left(\frac{e^A + e^{-A}}{2}\right)^2 - \left(\frac{e^A - e^{-A}}{2}\right)^2$

$\equiv \left(\frac{e^{2A} + 2 + e^{-2A}}{4}\right) - \left(\frac{e^{2A} - 2 + e^{-2A}}{4}\right)$

$\equiv \frac{4}{4} \equiv 1 \equiv \text{RHS}$

Use the definitions of cosh and sinh.

$e^A \times e^{-A} = \dfrac{e^A}{e^A} = 1.$

- **$\cosh^2 A - \sinh^2 A \equiv 1$**

Example 9

Prove that $\sinh(A + B) \equiv \sinh A \cosh B + \cosh A \sinh B$

$$\text{RHS} \equiv \sinh A \cosh B + \cosh A \sinh B$$

$$\equiv \left(\frac{e^{A} - e^{-A}}{2}\right)\left(\frac{e^{B} + e^{-B}}{2}\right) + \left(\frac{e^{A} + e^{-A}}{2}\right)\left(\frac{e^{B} - e^{-B}}{2}\right)$$

$$\equiv \left(\frac{e^{A+B} + e^{A-B} - e^{-A+B} - e^{-A-B}}{4}\right) + \left(\frac{e^{A+B} - e^{A-B} + e^{-A+B} - e^{-A-B}}{4}\right)$$

$$\equiv \frac{2e^{A+B} - 2e^{-A-B}}{4} \equiv \frac{e^{A+B} - e^{-(A+B)}}{2} \equiv \sinh(A + B) \equiv \text{LHS}$$

You can prove other sinh and cosh addition formulae similarly, giving:

- $\mathbf{\sinh(A \pm B) \equiv \sinh A \cosh B \pm \cosh A \sinh B}$
- $\mathbf{\cosh(A \pm B) \equiv \cosh A \cosh B \pm \sinh A \sinh B}$

Example 10

Prove that $\cosh 2A \equiv 1 + 2\sinh^2 A$

$$\text{RHS} \equiv 1 + 2\sinh^2 A$$

$$\equiv 1 + 2\left(\frac{e^{A} - e^{-A}}{2}\right)\left(\frac{e^{A} - e^{-A}}{2}\right)$$

Use the definition of sinh.

$$\equiv 1 + 2\left(\frac{e^{2A} - 2 + e^{-2A}}{4}\right) \equiv 1 - 1 + \left(\frac{e^{2A} + e^{-2A}}{2}\right)$$

$$\equiv \cosh 2A \equiv \text{LHS}$$

Given a trigonometric identity, it is generally possible to write down the corresponding hyperbolic identity using what is known as **Osborn's rule**:

- Replace cos by cosh: $\cos A \rightarrow \cosh A$
- Replace sin by sinh: $\sin A \rightarrow \sinh A$

However ...

- replace any product (or implied product) of two sin terms by **minus** the product of two sinh terms:

e.g. $\sin A \sin B \rightarrow -\sinh A \sinh B$

$\tan^2 A \rightarrow -\tanh^2 A$

This is the implied product of two sin terms because $\tan^2 A \equiv \dfrac{\sin^2 A}{\cos^2 A}$

Example 11

Write down the hyperbolic identity corresponding to:

a $\cos 2A \equiv 2\cos^2 A - 1$ **b** $\tan(A - B) \equiv \dfrac{\tan A - \tan B}{1 + \tan A \tan B}$

a $\cosh 2A \equiv 2\cosh^2 A - 1$

b $\tanh(A - B) \equiv \dfrac{\tanh A - \tanh B}{1 - \tanh A \tanh B}$

Implied product of two sin terms because $\tan A \tan B \equiv \dfrac{\sin A \sin B}{\cos A \cos B}$

Example 12

Given that $\sinh x = \frac{3}{4}$, find the exact value of:

a $\cosh x$ **b** $\tanh x$ **c** $\sinh 2x$

a Using $\cosh^2 x - \sinh^2 x \equiv 1$,

$\cosh^2 x - \frac{9}{16} = 1 \Rightarrow \cosh^2 x = \frac{25}{16}$

$\Rightarrow \cosh x = \frac{5}{4}$

b Using $\tanh x \equiv \dfrac{\sinh x}{\cosh x}$

$\tanh x = \frac{3}{4} \div \frac{5}{4} = \frac{3}{5}$

c Using $\sinh 2x \equiv 2\sinh x \cosh x$,

$\sinh 2x = 2 \times \frac{3}{4} \times \frac{5}{4} = \frac{15}{8}$

$\cosh x \geqslant 1$, so $\cosh x = -\frac{5}{4}$ is not possible.

You can solve equations involving hyperbolic functions.

Example 13

Solve $6\sinh x - 2\cosh x = 7$ for real values of x.

$$6\left(\frac{e^x - e^{-x}}{2}\right) - 2\left(\frac{e^x + e^{-x}}{2}\right) = 7$$

$$3e^x - 3e^{-x} - e^x - e^{-x} = 7$$

$$2e^x - 7 - 4e^{-x} = 0$$

$$2e^{2x} - 7e^x - 4 = 0$$

$$(2e^x + 1)(e^x - 4) = 0$$

$e^x = -\frac{1}{2}$, $e^x = 4$

$e^x = 4$

$x = \ln 4$

There is no hyperbolic identity that will easily transform the equation into an equation in just one hyperbolic function, so use the basic definitions.

There are no real values of x for which $e^x = -\frac{1}{2}$

Example 14

Solve $2\cosh^2 x - 5\sinh x = 5$, giving your answers as natural logarithms.

Using $\cosh^2 x - \sinh^2 x \equiv 1$,

Use this identity to transform the equation into an equation in just one hyperbolic function.

$2(1 + \sinh^2 x) - 5\sinh x = 5$

$2\sinh^2 x - 5\sinh x - 3 = 0$

$(2\sinh x + 1)(\sinh x - 3) = 0$

So $\sinh x = -\frac{1}{2}$ or $\sinh x = 3$

Then,

$x = \operatorname{arsinh}\left(-\frac{1}{2}\right)$ or $x = \operatorname{arsinh} 3$

$\Rightarrow x = \ln\left(-\frac{1}{2} + \sqrt{\frac{1}{4} + 1}\right)$ or $x = \ln(3 + \sqrt{9 + 1})$

Use $\operatorname{arsinh} x = \ln(x + \sqrt{x^2 + 1})$.

$\Rightarrow x = \ln\left(-\frac{1}{2} + \frac{\sqrt{5}}{2}\right)$ or $x = \ln(3 + \sqrt{10})$

Example 15

Solve $\cosh 2x - 5\cosh x + 4 = 0$, giving your answers as natural logarithms where appropriate.

Using $\cosh 2x \equiv 2\cosh^2 x - 1$,

Use this identity to transform the equation into an equation in just one hyperbolic function.

$2\cosh^2 x - 1 - 5\cosh x + 4 = 0$

$2\cosh^2 x - 5\cosh x + 3 = 0$

$(2\cosh x - 3)(\cosh x - 1) = 0$

So $\cosh x = \frac{3}{2}$ or $\cosh x = 1$

$\Rightarrow x = \ln\left(\frac{3}{2} \pm \sqrt{\frac{9}{4} - 1}\right)$ or $x = 0$

You can use $\operatorname{arcosh} x = \ln(x + \sqrt{x^2 - 1})$, but remember that both $\ln(x + \sqrt{x^2 - 1})$ and $\ln(x - \sqrt{x^2 - 1})$ are possible.

For any value of k greater than 1, $\cosh x = k$ will give two values of x, one positive and one negative.

$\Rightarrow x = \ln\left(\frac{3}{2} \pm \frac{\sqrt{5}}{2}\right)$ or $x = 0$

Exercise 6C

1 Prove the following identities, using the definitions of $\sinh x$ and $\cosh x$.

a $\sinh 2A \equiv 2\sinh A \cosh A$

b $\cosh(A - B) \equiv \cosh A \cosh B - \sinh A \sinh B$

c $\cosh 3A \equiv 4\cosh^3 A - 3\cosh A$

d $\sinh A - \sinh B \equiv 2\sinh\left(\frac{A - B}{2}\right)\cosh\left(\frac{A + B}{2}\right)$

2 Use Osborn's rule to write down the hyperbolic identities corresponding to the following trigonometric identities.

a $\sin(A - B) \equiv \sin A \cos B - \cos A \sin B$

b $\sin 3A \equiv 3\sin A - 4\sin^3 A$

c $\cos A + \cos B \equiv 2\cos\left(\frac{A + B}{2}\right)\cos\left(\frac{A - B}{2}\right)$

d $\cos 2A \equiv \frac{1 - \tan^2 A}{1 + \tan^2 A}$

e $\cos 2A \equiv \cos^4 A - \sin^4 A$

3 Given that $\cosh x = 2$, find the exact values of:

a $\sinh x$ **b** $\tanh x$ **c** $\cosh 2x$

4 Given that $\sinh x = -1$, find the exact values of:

a $\cosh x$ **b** $\sinh 2x$ **c** $\tanh 2x$

5 Solve the following equations, giving your answers as natural logarithms.

a $3\sinh x + 4\cosh x = 4$ **b** $7\sinh x - 5\cosh x = 1$ **c** $30\cosh x = 15 + 26\sinh x$

d $13\sinh x - 7\cosh x + 1 = 0$ **e** $\cosh 2x - 5\sinh x = 13$ **f** $3\sinh^2 x - 13\cosh x + 7 = 0$

g $\sinh 2x - 7\sinh x = 0$ **h** $4\cosh x + 13e^{-x} = 11$ **i** $2\tanh x = \cosh x$

(E) **6 a** Starting from the definitions of $\sinh x$ and $\cosh x$ in terms of exponentials, prove that

$$\cosh 2x \equiv 2\cosh^2 x - 1$$ **(3 marks)**

b Solve the equation

$$\cosh 2x - 3\cosh x = 8$$

giving your answers as exact logarithms. **(5 marks)**

(E) **7** Solve the equation

$$2\sinh^2 x - 5\cosh x = 5$$

giving your answer in terms of natural logarithms in simplest form. **(6 marks)**

(P) **8** Joshua is asked to prove the following identity:

$$\frac{1 + \tanh^2 x}{1 - \tanh^2 x} \equiv 2\cosh^2 x - 1$$

His answer is below.

$$\frac{1 + \tanh^2 x}{1 - \tanh^2 x} \equiv \frac{\operatorname{sech}^2 x}{2 - \operatorname{sech}^2 x}$$ (using $\operatorname{sech}^2 x \equiv 1 + \tanh^2 x$: same identity as the trig one)

$$\equiv \frac{\operatorname{sech}^2 x}{2} - 1$$ (splitting the fraction up and cancelling)

$$\equiv \frac{2}{\operatorname{sech}^2 x} - 1$$ (taking the reciprocal of both terms)

$$\equiv 2\cosh^2 x - 1$$

Joshua has made three errors. Explain the errors and provide a correct proof. **(6 marks)**

(E/P) **9 a** Express $10\cosh x + 6\sinh x$ in the form $R\cosh(x + a)$ where $R > 0$. Give the value of a correct to 3 decimal places. **(4 marks)**

Hint Use the identity for $\cosh(A + B)$.

b Write down the minimum value of $10\cosh x + 6\sinh x$. **(1 mark)**

c Use your answer to part **a** to solve the equation $10\cosh x + 6\sinh x = 11$. Give your answers to 3 decimal places. **(4 marks)**

6.4 Differentiating hyperbolic functions

You can differentiate hyperbolic functions.

- $\frac{d}{dx}(\sinh x) = \cosh x$
- $\frac{d}{dx}(\cosh x) = \sinh x$
- $\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$

Watch out The rules for $\sinh x$ and $\tanh x$ are the same as the corresponding rules for $\sin x$ and $\tan x$. However, the derivative of $\cosh x$ is **positive** $\sinh x$.

Example 16

Show that $\frac{d}{dx}(\cosh x) = \sinh x$.

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

Use the definition of $\cosh x$.

$$\text{So } \frac{d}{dx}(\cosh x) = \frac{d}{dx}\left(\frac{e^x + e^{-x}}{2}\right)$$

$$= \frac{e^x - e^{-x}}{2}$$

Differentiate with respect to x:
$\frac{d}{dx}(e^{-x}) = -e^{-x}$

$$\frac{e^x - e^{-x}}{2} = \sinh x$$

By definition.

$$\text{So } \frac{d}{dx}(\cosh x) = \sinh x$$

Example 17

Differentiate $\cosh 3x$ with respect to x.

$$\frac{d}{dx}(\cosh 3x) = 3\sinh 3x$$

Use the chain rule.

Example 18

Differentiate $x^2 \cosh 4x$ with respect to x.

$$\frac{d}{dx}(x^2 \cosh 4x) = \frac{d}{dx}(x^2)\cosh 4x + x^2 \frac{d}{dx}(\cosh 4x)$$

Use the product rule.

$$= 2x \cosh 4x + x^2 \times 4 \sinh 4x$$

$$= 2x \cosh 4x + 4x^2 \sinh 4x$$

Example 19

Given that $y = A\cosh 3x + B\sinh 3x$, where A and B are constants, prove that $\frac{d^2y}{dx^2} = 9y$.

$\frac{dy}{dx} = 3A\sinh 3x + 3B\cosh 3x$ — Differentiate y.

$\frac{d^2y}{dx^2} = 9A\cosh 3x + 9B\sinh 3x$ — Differentiate again.

$= 9(A\cosh 3x + B\sinh 3x)$ — Factorise.

$= 9y$ — $y = A\cosh 3x + B\sinh 3x$

You can also differentiate the inverse hyperbolic functions.

- $\frac{d}{dx}(\text{arsinh}\, x) = \frac{1}{\sqrt{x^2+1}}$
- $\frac{d}{dx}(\text{arcosh}\, x) = \frac{1}{\sqrt{x^2-1}}, x > 1$
- $\frac{d}{dx}(\text{artanh}\, x) = \frac{1}{1-x^2}, |x| < 1$

Example 20

Given $y = x\,\text{arcosh}\, x$, find $\frac{dy}{dx}$

$\frac{dy}{dx} = \text{arcosh}\, x + x \times \frac{1}{\sqrt{x^2-1}}$ — Use the product rule.

$= \text{arcosh}\, x + \frac{x}{\sqrt{x^2-1}}$

Example 21

Given $y = (\text{arcosh}\, x)^2$, prove that $(x^2-1)\left(\frac{dy}{dx}\right)^2 = 4y$.

$\frac{dy}{dx} = 2\text{arcosh}\, x \times \frac{1}{\sqrt{x^2-1}}$ — Use the chain rule.

$\Rightarrow \sqrt{x^2-1}\frac{dy}{dx} = 2\text{arcosh}\, x$ — Multiply by $\sqrt{x^2-1}$.

$\Rightarrow (x^2-1)\left(\frac{dy}{dx}\right)^2 = 4(\text{arcosh}\, x)^2$ — Square both sides.

But $y = (\text{arcosh}\, x)^2$

so $(x^2-1)\left(\frac{dy}{dx}\right)^2 = 4y$

Example 22

a Show that $\frac{d}{dx}(\text{arsinh}\, x) = \frac{1}{\sqrt{1+x^2}}$

b Find the first two non-zero terms of the series expansion of arsinh x.

The general form for the series expansion of arsinh x is given by

$$\text{arsinh}\, x = \sum_{r=0}^{\infty}\left(\frac{(-1)^n(2n)!}{2^{2n}(n!)^2}\right)\frac{x^{2n+1}}{2n+1}$$

c Find, in simplest terms, the coefficient of x^5.

d Use your approximation up to and including the term in x^5 to find an approximate value for arsinh 0.5.

e Calculate the percentage error in using this approximation.

a Let $y = \text{arsinh}\, x$

then $\sinh y = x$

$\Rightarrow \frac{dx}{dy} = \cosh y$ — Differentiate.

$= \sqrt{\sinh^2 y + 1}$ — Use $\cosh^2 x - \sinh^2 x \equiv 1$.

But $\sinh y = x$,

so $\frac{dx}{dy} = \sqrt{x^2+1}$ — Take the reciprocal of both sides.

therefore $\frac{dy}{dx} = \frac{1}{\sqrt{x^2+1}}$

b $f(x) = \text{arsinh}\, x \Rightarrow f(0) = 0$

$f'(x) = \frac{1}{\sqrt{1+x^2}} \Rightarrow f'(0) = 1$

$f''(x) = -\frac{x}{(1+x^2)^{\frac{3}{2}}} \Rightarrow f''(0) = 0$ — Find the second derivative using the chain rule or quotient rule.

Watch out You need to find the first two non-zero terms. $f(0) = 0$ and $f''(0) = 0$ so there is no constant term and no x^2 term. Differentiate again to find the x^3 term.

$f'''(x) = \frac{2x^2-1}{(1+x^2)^{\frac{5}{2}}} \Rightarrow f'''(0) = -1$ — Find the third derivative using the quotient rule.

$\text{arsinh}\, x \approx x - \frac{x^3}{3!} = x - \frac{1}{6}x^3$ — Use the standard Maclaurin series expansion.

← Chapter 2

c The x^5 term will be when $n = 2$:

$\left(\frac{(-1)^2 4!}{2^4(2!)^2}\right)\frac{x^5}{5} = \frac{1 \times 24}{16 \times 4} \times \frac{x^5}{5} = \frac{3}{40}x^5$

The coefficient of x^5 is $\frac{3}{40}$

Problem-solving

The general term contains $(-1)^n$ and a power of x^{2n+1}. The non-zero terms will all have odd powers of x and will alternate signs.

d $\text{arsinh}\, 0.5 \approx 0.5 - \frac{1}{6}(0.5)^3 + \frac{3}{40}(0.5)^5$

$= 0.48151\ldots$

e $\%\ \text{error} = \frac{0.48151\ldots - \text{arsinh}\, 0.5}{\text{arsinh}\, 0.5} \times 100$

$= 0.062\%$ (3 d.p.)

Exercise 6D

1 Differentiate with respect to x:

a $\sinh 2x$ **b** $\cosh 5x$ **c** $\tanh 2x$ **d** $\sinh 3x$

e $\coth 4x$ **f** $\operatorname{sech} 2x$ **g** $e^{-x}\sinh x$

h $x\cosh 3x$ **i** $\dfrac{\sinh x}{3x}$ **j** $x^2\cosh 3x$

k $\sinh 2x\cosh 3x$ **l** $\ln(\cosh x)$ **m** $\sinh x^3$

n $\cosh^2 2x$ **o** $e^{\cosh x}$ **p** $\operatorname{cosech} x$

Hint For part **e**, use $\coth x = \dfrac{1}{\tanh x}$

For part **f**, use $\operatorname{sech} 2x = \dfrac{1}{\cosh 2x}$

For part **p**, use $\operatorname{cosech} x = \dfrac{1}{\sinh x}$

(E) **2** If $y = a\cosh nx + b\sinh nx$, where a and b are constants, prove that $\dfrac{d^2y}{dx^2} = n^2y$. **(4 marks)**

(E/P) **3** Find the exact coordinates of the stationary point on the curve with equation $y = 12\cosh x - \sinh x$. **(7 marks)**

(E) **4** Given that $y = \cosh 3x \sinh x$, find $\dfrac{d^2y}{dx^2}$ **(6 marks)**

5 Differentiate:

a $\operatorname{arcosh} 2x$ **b** $\operatorname{arsinh}(x+1)$ **c** $\operatorname{artanh} 3x$

d $\operatorname{arsech} x$ **e** $\operatorname{arcosh} x^2$ **f** $\operatorname{arcosh} 3x$

g $x^2\operatorname{arcosh} x$ **h** $\operatorname{arsinh}\dfrac{x}{2}$ **i** $e^{x^3}\operatorname{arsinh} x$

j $\operatorname{arsinh} x\operatorname{arcosh} x$ **k** $\operatorname{arcosh} x\operatorname{sech} x$ **l** $x\operatorname{arcosh} 3x$

(P) **6** Prove that:

a $\dfrac{d}{dx}(\operatorname{arcosh} x) = \dfrac{1}{\sqrt{x^2-1}}$, $x > 1$ **b** $\dfrac{d}{dx}(\operatorname{artanh} x) = \dfrac{1}{1-x^2}$, $|x| < 1$

(E) **7** Given that $y = \operatorname{artanh}\left(\dfrac{e^x}{2}\right)$, prove that

$$(4 - e^{2x})\frac{dy}{dx} = 2e^x$$ **(6 marks)**

(E) **8** Given that $y = \operatorname{arsinh} x$, show that

$$(1+x^2)\frac{d^3y}{dx^3} + 3x\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$$ **(7 marks)**

(E) **9** If $y = (\operatorname{arcosh} x)^2$, find $\dfrac{d^2y}{dx^2}$ **(6 marks)**

E **10** Find the equation of the tangent at the point where $x = \frac{12}{13}$ on the curve with equation $y = \text{artanh}\, x$. **(3 marks)**

E/P **11** Show that the equation of the normal at the point where $x = 2$ on the curve with equation $y = \text{arcosh}\, 2x$ can be written as $y = ax + b + \ln c$, where a, b and c are exact real numbers to be found. **(5 marks)**

E/P **12** **a** Find the first three non-zero terms of the Maclaurin series for $\cosh x$. **(5 marks)**

b Hence find the percentage error when this approximation is used to evaluate $\cosh 0.2$. **(3 marks)**

E/P **13** **a** Find the first three non-zero terms of the Maclaurin series for $\sinh x$. **(5 marks)**

b Hence find an expression for the nth non-zero term of the Maclaurin series for $\sinh x$. **(2 marks)**

E/P **14** $y = \tanh x$

a Use a Maclaurin series expansion to show that $y \approx x - \frac{1}{3}x^3$. **(5 marks)**

b Find the percentage error when the approximation in **a** is used to evaluate $\tanh 0.8$. **(2 marks)**

E/P **15** $y = \text{artanh}\, x$

a Show that the first three non-zero terms of the Maclaurin series of y are $x + \frac{1}{3}x^3 + \frac{1}{5}x^5$ **(6 marks)**

b Hence write down the general term for the nth non-zero term in the series expansion. **(1 mark)**

c Find the first two non-zero terms of the series expansion of $\cosh x\, \text{artanh}\, x$. **(5 marks)**

E/P **16** $y = \sinh x \cosh 2x$

Find the first three non-zero terms of the Maclaurin series for y, giving each coefficient in its simplest form. **(7 marks)**

E/P **17** $y = \cos x \cosh x$

a Show that $\frac{\mathrm{d}^4y}{\mathrm{d}x^4} = -4y$. **(4 marks)**

b Hence find the first three non-zero terms of the Maclaurin series for y, giving each coefficient in its simplest form. **(4 marks)**

c Hence write the series expansion for y in the form $\sum_{r=0}^{\infty} \mathrm{f}(r)x^{4r}$, where f is function to be determined. **(2 marks)**

Challenge

Find the first three non-zero terms of the series expansion of $\text{sech}\, x$.

6.5 Integrating hyperbolic functions

You can integrate hyperbolic functions.

- **You should be familiar with the following integrals:**
  - $\int \sinh x \, dx = \cosh x + c$ since $\frac{d}{dx}(\cosh x) = \sinh x$
  - $\int \cosh x \, dx = \sinh x + c$ since $\frac{d}{dx}(\sinh x) = \cosh x$
  - $\int \frac{1}{\sqrt{1+x^2}} dx = \text{arsinh } x + c$ since $\frac{d}{dx}(\text{arsinh } x) = \frac{1}{\sqrt{1+x^2}}$
  - $\int \frac{1}{\sqrt{x^2-1}} dx = \text{arcosh } x + c, x > 1$ since $\frac{d}{dx}(\text{arcosh } x) = \frac{1}{\sqrt{x^2-1}}, x > 1$

Example 23

Find $\int \cosh(4x - 1)\, dx$.

$\int \cosh(4x - 1)dx = \frac{1}{4}\sinh(4x - 1) + c$

Example 24

Find $\int \frac{2 + 5x}{\sqrt{x^2 + 1}} dx$.

$$\int \frac{2 + 5x}{\sqrt{x^2 + 1}} dx = \int \frac{2}{\sqrt{x^2 + 1}} dx + \int \frac{5x}{\sqrt{x^2 + 1}} dx$$

$$= 2\int \frac{1}{\sqrt{x^2 + 1}} dx + 5\int x(1 + x^2)^{-\frac{1}{2}} dx$$

$$= 2\,\text{arsinh } x + 5\sqrt{1 + x^2} + c$$

Splitting the numerator gives two recognisable integrals.

This is the standard result for arsinh x.

This integral is of the form $k\int f'(x)(f(x))^n dx$.
← Pure Year 2, Chapter 9

Example 25

Find:

a $\int \cosh^5 2x \sinh 2x \, dx$ **b** $\int \tanh x \, dx$

a $\int \cosh^5 2x \sinh 2x \, dx = \frac{1}{2}\int (\cosh 2x)^5 (2 \sinh 2x)\, dx$

$= \frac{1}{12}\cosh^6 2x + c$

Use $\int (f(x))^n f'(x)\, dx = \frac{(f(x))^{n+1}}{n+1} + c$, with $f(x) = \cosh 2x$ and $n = 5$.

b $\int \tanh x \, dx = \int \frac{\sinh x}{\cosh x} \, dx$

$= \ln \cosh x + c$

Use $\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + c$, with $f(x) = \cosh x$. Modulus signs are not necessary because $\cosh x > 0$, for all x.

- $\int \tanh x \, dx = \ln \cosh x + c$

Example 26

Find the following integrals.

a $\int \cosh^2 3x \, dx$ b $\int \sinh^3 x \, dx$

a $\int \cosh^2 3x \, dx = \int \frac{1 + \cosh 6x}{2} dx$

$= \frac{1}{2}\left(x + \frac{\sinh 6x}{6}\right) + c$

$= \frac{1}{2}x + \frac{1}{12}\sinh 6x + c$

Use $\cosh 2A \equiv 2\cosh^2 A - 1$ with $A \equiv 3x$.

b $\int \sinh^3 x \, dx = \int \sinh^2 x \sinh x \, dx$

$= \int (\cosh^2 x - 1)\sinh x \, dx$

$= \int \cosh^2 x \sinh x \, dx - \int \sinh x \, dx$

$= \frac{1}{3}\cosh^3 x - \cosh x + c$

Problem-solving

For small odd values of n, you can use $\int \sinh^n x \, dx = \int \sinh^{n-1} x \sinh x \, dx$.

$\int \cosh^n x \, dx$, for odd values of n, can be found similarly.

Sometimes the exponential definition of a hyperbolic function can be used to find a given integral.

Example 27

Find $\int e^{2x} \sinh x \, dx$.

$\int e^{2x} \sinh x \, dx = \int e^{2x}\left(\frac{e^x - e^{-x}}{2}\right) dx$

$= \frac{1}{2}\int (e^{3x} - e^x) \, dx$

$= \frac{1}{2}\left(\frac{e^{3x}}{3} - e^x\right) + c$

$= \frac{1}{6}(e^{3x} - 3e^x) + c$

Use the definition of $\sinh x$.

You need to be able to use hyperbolic substitutions to find the integrals of expressions of the forms

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx \quad \text{and} \quad \int \frac{1}{\sqrt{x^2 + a^2}} dx$$

Example 28

By using an appropriate substitution, find $\int \frac{1}{\sqrt{x^2 - a^2}} \, dx$, $x > a$.

As $\cosh^2 u - 1 \equiv \sinh^2 u$, it follows that

$a^2 \cosh^2 u - a^2 = a^2 \sinh^2 u$

Use the substitution $x = a \cosh u$,

This is similar to using the substitution $x = a \sin u$ to integrate $\frac{1}{\sqrt{a^2 - x^2}}$ ← Section 3.4

$\frac{dx}{du} = a \sinh u$

so dx can be replaced by $a \sinh u \, du$.

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \int \frac{1}{\sqrt{a^2 \cosh^2 u - a^2}} a \sinh u \, du$$

$$= \int \frac{1}{\cancel{a}\sinh u} \cancel{a} \sinh u \, du$$

$$= u + c$$

$$= \operatorname{arcosh}\left(\frac{x}{a}\right) + c$$

$x = a \cosh u \Rightarrow \cosh u = \frac{x}{a} \Rightarrow u = \operatorname{arcosh}\left(\frac{x}{a}\right)$

- $\int \frac{1}{\sqrt{a^2 + x^2}} dx = \operatorname{arsinh}\left(\frac{x}{a}\right) + c$
- $\int \frac{1}{\sqrt{x^2 - a^2}} dx = \operatorname{arcosh}\left(\frac{x}{a}\right) + c, \; x > a$

Example 29

Show that $\int_5^8 \frac{1}{\sqrt{x^2 - 16}} dx = \ln\left(\frac{2 + \sqrt{3}}{2}\right)$

$$\int_5^8 \frac{1}{\sqrt{x^2 - 16}} dx = \left[\operatorname{arcosh}\left(\frac{x}{4}\right)\right]_5^8$$

Use the result for $\int \frac{1}{\sqrt{x^2 - a^2}} dx$ with $a = 4$.

$$= \operatorname{arcosh} 2 - \operatorname{arcosh}\left(\frac{5}{4}\right)$$

$$= \ln(2 + \sqrt{3}) - \ln\left(\frac{5}{4} + \sqrt{\frac{9}{16}}\right)$$

Use $\operatorname{arcosh} x = \ln (x + \sqrt{x^2 - 1})$.

$$= \ln(2 + \sqrt{3}) - \ln 2$$

$$= \ln\left(\frac{2 + \sqrt{3}}{2}\right)$$

Use $\ln a - \ln b = \ln\left(\frac{a}{b}\right)$

Example 30

Show that $\int \sqrt{1+x^2}\,\mathrm{d}x = \frac{1}{2}\operatorname{arsinh} x + \frac{1}{2}x\sqrt{1+x^2} + c$.

$x = \sinh u \Rightarrow \frac{dx}{du} = \cosh u$

so dx can be replaced by $\cosh u\,du$

$$\int \sqrt{1+x^2}\,dx = \int \sqrt{1+\sinh^2 u}\cosh u\,du$$

$$= \int \cosh^2 u\,du$$

$$= \frac{1}{2}\int (1+\cosh 2u)\,du$$

Use $\cosh 2u \equiv 2\cosh^2 u - 1$.

$$= \frac{1}{2}\left(u + \frac{\sinh 2u}{2}\right) + c$$

You need to be able to use $x = \sinh u$, so use $\sinh 2u \equiv 2\sinh u \cosh u$.

$$= \frac{1}{2}(u + \sinh u \cosh u) + c$$

$$= \frac{1}{2}\operatorname{arsinh} x + \frac{1}{2}x\sqrt{1+x^2} + c$$

$u = \operatorname{arsinh} x$ and $\cosh u \equiv \sqrt{1+\sinh^2 u}$.

Example 31

By using a hyperbolic substitution, evaluate $\int_0^6 \frac{x^3}{\sqrt{x^2+9}}\,\mathrm{d}x$.

Use the substitution $x = 3\sinh u$

You need to reduce $x^2 + 9$ to a single term. Using $x = 3\sinh u$ gives $9\sinh^2 u + 9 = 9(\sinh^2 u + 1) = 9\cosh^2 u$

$\frac{dx}{du} = 3\cosh u$

So dx can be replaced by $3\cosh u\,du$

$$\int_0^6 \frac{x^3}{\sqrt{x^2+9}}\,dx = \int_0^{\operatorname{arsinh}2} \frac{27\sinh^3 u}{3\cosh u} 3\cosh u\,du$$

When $x = 6$, $\sinh u = 2 \Rightarrow u = \operatorname{arsinh} 2$.
When $x = 0$, $\sinh u = 0 \Rightarrow u = 0$.

$$= 27\int_0^{\operatorname{arsinh}2} \sinh^3 u\,du$$

Use the method from Example 26.

$$= 27\left[\tfrac{1}{3}\cosh^3 u - \cosh u\right]_0^{\operatorname{arsinh}2}$$

$$= 27\left(\frac{5\sqrt{5}}{3} - \sqrt{5}\right) - 27\left(\tfrac{1}{3} - 1\right)$$

As $\sinh u = 2$, $\cosh u = \sqrt{1+2^2} = \sqrt{5}$.

$$= 18\sqrt{5} + 18$$

Example 32

Find $\int \frac{1}{\sqrt{12x + 2x^2}}\,\mathrm{d}x$.

$12x + 2x^2 = 2(x^2 + 6x)$
$= 2((x + 3)^2 - 9)$

Complete the square.

So $\int \frac{1}{\sqrt{12x + 2x^2}}dx = \int \frac{1}{\sqrt{2((x + 3)^2 - 9)}}dx$

Let $u = x + 3$, then $du = dx$

Choose the substitution.

$\int \frac{1}{\sqrt{12x + 2x^2}}dx = \frac{1}{\sqrt{2}}\int \frac{1}{\sqrt{u^2 - 9}}du$

$= \frac{1}{\sqrt{2}}\operatorname{arcosh}\left(\frac{u}{3}\right) + c$

Select the standard form.

$= \frac{1}{\sqrt{2}}\operatorname{arcosh}\left(\frac{x + 3}{3}\right) + c$

Rewrite the answer in terms of x.

Example 33

Use the substitution $x = \frac{1}{2}(3 + 4\cosh u)$ to find $\int \frac{1}{\sqrt{4x^2 - 12x - 7}}\,\mathrm{d}x$.

$x = \frac{1}{2}(3 + 4\cosh u) \Rightarrow \frac{dx}{du} = \frac{1}{2}(4\sinh u) = 2\sinh u$

so dx can be replaced by $2\sinh u\,du$

$4x^2 - 12x - 7 = 4\left(\frac{1}{2}(3 + 4\cosh u)\right)^2 - 6(3 + 4\cosh u) - 7$
$= 9 + 24\cosh u + 16\cosh^2 u - 18 - 24\cosh u - 7$
$= 16\cosh^2 u - 16$
$= 16\sinh^2 u$

Use $\cosh^2 u - \sinh^2 u \equiv 1$

So $\int \frac{1}{\sqrt{4x^2 - 12x - 7}}dx = \int \frac{1}{4\sinh u} \times 2\sinh u\,du$

$= \frac{1}{2}\int 1du$
$= \frac{1}{2}u + c$
$= \frac{1}{2}\operatorname{arcosh}\left(\frac{2x - 3}{4}\right) + c$

As $x = \frac{1}{2}(3 + 4\cosh u)$, $\cosh u = \left(\frac{2x - 3}{4}\right)$

Exercise 6E

1 Integrate the following with respect to x.

a $\sinh x + 3\cosh x$ **b** $\cosh x - \frac{1}{\cosh^2 x}$ **c** $\frac{\sinh x}{\cosh^2 x}$

2 Find:

a $\int \sinh 2x \, dx$ **b** $\int \cosh\left(\frac{x}{3}\right) dx$

3 Find:

a $\int \frac{1+x}{\sqrt{x^2-1}} dx$ **b** $\int \frac{x-3}{\sqrt{1+x^2}} dx$

4 Find:

a $\int \sinh^3 x \cosh x \, dx$ **b** $\int \tanh 4x \, dx$ **c** $\int \sqrt{\cosh 2x} \sinh 2x \, dx$

5 Find:

a $\int \frac{\sinh x}{2+3\cosh x} dx$ **b** $\int \frac{1+\tanh x}{\cosh^2 x} dx$ **c** $\int \frac{5\cosh x + 2\sinh x}{\cosh x} dx$

6 Use integration by parts to find $\int x \sinh 3x \, dx$.

7 Find:

a $\int e^x \cosh x \, dx$ **b** $\int e^{-2x} \sinh 3x \, dx$ **c** $\int \cosh x \cosh 3x \, dx$

8 Evaluate $\int_0^1 \frac{1}{\sinh x + \cosh x} dx$, giving your answer in terms of e.

9 Use appropriate identities to find:

a $\int \sinh^2 x \, dx$ **b** $\int \sinh^2 x \cosh^2 x \, dx$ **c** $\int \cosh^5 x \, dx$

(E) **10** Show that $\int_0^{\ln 2} \cosh^2\left(\frac{x}{2}\right) dx = \frac{1}{8}(3 + \ln 16)$. **(7 marks)**

11 Use suitable substitutions to find:

a $\int \frac{1}{\sqrt{x^2-9}} dx$ **b** $\int \frac{1}{\sqrt{4x^2+25}} dx$

12 Write down the results for the following:

a $\int \frac{3}{\sqrt{x^2+9}} dx$ **b** $\int \frac{1}{\sqrt{x^2-2}} dx$

13 Find:

a $\int \frac{1}{\sqrt{4x^2 - 12}} \mathrm{d}x$ **b** $\int \frac{1}{\sqrt{9x^2 + 16}} \mathrm{d}x$

14 Evaluate $\int_1^2 \frac{3}{\sqrt{1 + 4x^2}} \mathrm{d}x$.

15 Evaluate, giving your answers as a single natural logarithms.

a $\int_0^4 \frac{1}{\sqrt{x^2 + 16}} \mathrm{d}x$ **b** $\int_{13}^{15} \frac{1}{\sqrt{x^2 - 144}} \mathrm{d}x$

(E) **16** Use the substitution $x = \sinh^2 u$ to find $\int \sqrt{\frac{x}{x+1}} \mathrm{d}x$, $x > 0$. **(6 marks)**

(E) **17** By using the substitution $u = x^2$, evaluate $\int_2^3 \frac{2x}{\sqrt{x^4 - 1}} \mathrm{d}x$. **(6 marks)**

(E) **18** Use the substitution $x = 2\cosh u$ to show that

$\int \sqrt{x^2 - 4} \, \mathrm{d}x = \frac{1}{2}x\sqrt{x^2 - 4} - 2\operatorname{arcosh}\left(\frac{x}{2}\right) + c$ **(6 marks)**

(E/P) **19** **a** Show that $\int \frac{1}{2\cosh x - \sinh x} \mathrm{d}x$ can be written as $\int \frac{2\mathrm{e}^x}{\mathrm{e}^{2x} + 3} \mathrm{d}x$. **(3 marks)**

b Hence, by using the substitution $u = \mathrm{e}^x$, find $\int \frac{1}{2\cosh x - \sinh x} \mathrm{d}x$. **(5 marks)**

(E/P) **20** Using the substitution $u = \frac{2}{3}\sinh x$, evaluate $\int_0^1 \frac{\cosh x}{\sqrt{4\sinh^2 x + 9}} \mathrm{d}x$. **(8 marks)**

(P) **21** Find the following:

a $\int \frac{1}{\sqrt{x^2 - 4x - 12}} \mathrm{d}x$ **b** $\int \frac{1}{\sqrt{x^2 + 6x + 10}} \mathrm{d}x$

c $\int \frac{1}{2x^2 + 4x + 7} \mathrm{d}x$ **d** $\int \frac{1}{\sqrt{9x^2 - 8x + 1}} \mathrm{d}x$

Problem-solving

Complete the square in the denominator to identify a suitable substitution.

$\frac{1}{\sqrt{x^2 - 4x - 12}} = \frac{1}{\sqrt{(x-2)^2 - 16}}$ so the substitution $u = x - 2$ will reduce the integral in part **a** to a standard form.

(P) **22** Find:

a $\int \frac{1}{\sqrt{4x^2 - 12x + 10}} \mathrm{d}x$ **b** $\int \frac{1}{\sqrt{4x^2 - 12x + 4}} \mathrm{d}x$

(P) **23** Evaluate $\int_0^1 \frac{1}{\sqrt{x^2 + 2x + 5}} \mathrm{d}x$

E/P **24** Evaluate $\int_1^3 \frac{1}{\sqrt{x^2 - 2x + 2}} \mathrm{d}x$, giving your answer as a single natural logarithm. **(8 marks)**

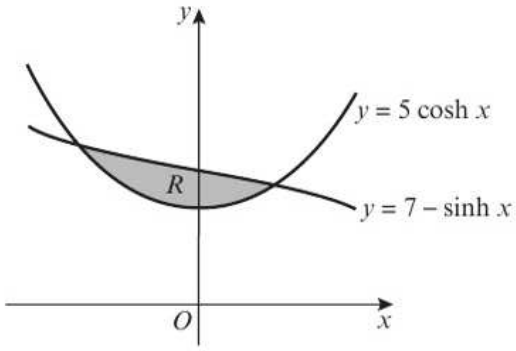
E/P **25** Show that $\int_1^3 \frac{1}{\sqrt{3x^2 - 6x + 7}} \mathrm{d}x = \frac{1}{\sqrt{3}} \ln(2 + \sqrt{3})$ **(8 marks)**

E/P **26** The curves shown have equations

$y = 5 \cosh x$ and $y = 7 - \sinh x$.

a Using the definitions of $\sinh x$ and $\cosh x$ in terms of e^x, find exact values for the x-coordinates of the two points where the curves intersect. **(6 marks)**

b Use calculus to find the area of the finite region, R, between the two curves, giving your answer in the form $\ln a + b$ where a and b are integers. **(6 marks)**



E/P **27** The region bounded by the curve $y = \sinh x$, the line $x = 1$ and the positive x-axis is rotated through 360° about the x-axis. Show that the volume of the solid of revolution formed is

$\frac{\pi}{8e^2}(e^4 - 4e^2 - 1)$ **(5 marks)**

Challenge

1 Using the substitution $x = 1 + \sinh\theta$, show that

$$\int \frac{1}{(x^2 - 2x + 2)^{\frac{3}{2}}} \mathrm{d}x = \frac{x - 1}{\sqrt{x^2 - 2x + 2}} + c$$

2 By means of a suitable substitution, or otherwise, find:

a $\int x \cosh^2(x^2) \mathrm{d}x$ **b** $\int \frac{x}{\cosh^2(x^2)} \mathrm{d}x$

Mixed exercise 6

1 Find the exact value of: **a** $\sinh(\ln 3)$ **b** $\cosh(\ln 5)$ **c** $\tanh(\ln \frac{1}{4})$

2 Given that $\operatorname{artanh} x - \operatorname{artanh} y = \ln 5$, find y in terms of x.

E/P **3** Using the definitions of $\sinh x$ and $\cosh x$, prove that

$\sinh(A - B) \equiv \sinh A \cosh B - \cosh A \sinh B$ **(5 marks)**

E/P **4** Using definitions in terms of exponentials, prove that

$$\sinh x \equiv \frac{2\tanh \frac{1}{2}x}{1 - \tanh^2 \frac{1}{2}x}$$ **(5 marks)**

E **5** Solve, giving your answers as natural logarithms

$9\cosh x - 5\sinh x = 15$ **(6 marks)**

E **6** Solve, giving your answers as natural logarithms

$23\sinh x - 17\cosh x + 7 = 0$ **(6 marks)**

E **7** Solve, giving your answers as natural logarithms

$3\cosh^2 x + 11\sinh x = 17$ **(6 marks)**

E **8** **a** On the same diagram, sketch the graphs of $y = 6 + \sinh x$ and $y = \sinh 3x$. **(2 marks)**

b Using the identity $\sinh 3x \equiv 3\sinh x + 4\sinh^3 x$, show that the graphs intersect where $\sinh x = 1$ and hence find the exact coordinates of the point of intersection. **(5 marks)**

E/P **9** **a** Given that $13\cosh x + 5\sinh x = R\cosh(x + \alpha)$, $R > 0$, use the identity $\cosh(A + B) \equiv \cosh A\cosh B + \sinh A\sinh B$ to find the values of R and α, giving the value of α to 3 decimal places. **(4 marks)**

b Write down the minimum value of $13\cosh x + 5\sinh x$. **(1 mark)**

E/P **10** **a** Express $3\cosh x + 5\sinh x$ in the form $R\sinh(x + \alpha)$, where $R > 0$. Give α to 3 decimal places. **(4 marks)**

b Use the answer to part **a** to solve the equation $3\cosh x + 5\sinh x = 8$, giving your answer to 2 decimal places. **(3 marks)**

c Solve $3\cosh x + 5\sinh x = 8$ by using the definitions of $\cosh x$ and $\sinh x$. **(4 marks)**

11 Given $y = \cosh 2x$, find $\frac{dy}{dx}$

12 Differentiate with respect to x:

a $\operatorname{arsinh} 3x$ **b** $\operatorname{arsinh}(x^2)$ **c** $\operatorname{arcosh}\frac{x}{2}$ **d** $x^2\operatorname{arcosh} 2x$

E **13** Given that $y = (\operatorname{arsinh} x)^2$, prove that

$$(1 + x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx} - 2 = 0$$ **(5 marks)**

E/P **14** Given that $f(x) = 5\cosh x - 3\sinh x$, find:

a $f'(x)$ **(2 marks)**

b the turning point on the curve $y = f(x)$ **(4 marks)**

E **15** Differentiate $\operatorname{arcosh}(\sinh 2x)$. **(4 marks)**

E/P **16** $y = \sin x \cosh x$

a Show that $\dfrac{\mathrm{d}^4 x}{\mathrm{d}y^4} = -4y$. **(4 marks)**

b Hence find the first three non-zero terms of the Maclaurin series for y, giving each coefficient in its simplest form. **(4 marks)**

E/P **17** $y = \sinh 2x \cosh x$

Find the first three non-zero terms of the Maclaurin series for y, giving each coefficient in its simplest form. **(7 marks)**

E/P **18** $4x^2 + 4x + 17 \equiv (ax + b)^2 + c, a > 0$.

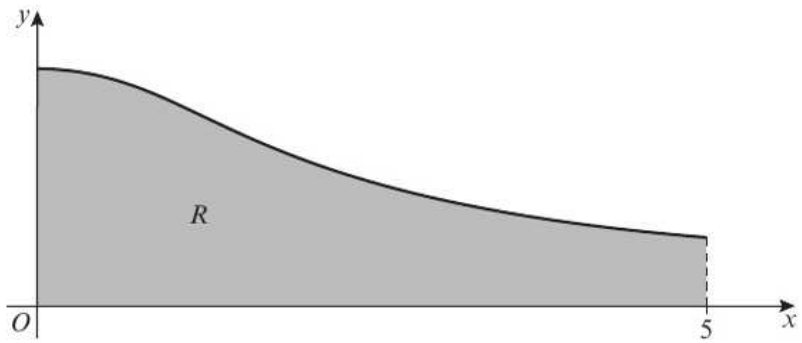
a Find the values of a, b and c. **(3 marks)**

b Find the exact value of $\displaystyle\int_{-\frac{1}{2}}^{\frac{3}{2}} \frac{1}{\sqrt{4x^2 + 4x + 17}}\,\mathrm{d}x$ **(6 marks)**

19 Find the following:

a $\displaystyle\int \sinh 4x \cosh 6x\,\mathrm{d}x$ **b** $\displaystyle\int \mathrm{e}^x \sinh x\,\mathrm{d}x$

E/P **20** The diagram shows the cross-section R of an artificial ski slope.



The slope is modelled by the curve with equation

$$y = \frac{10}{\sqrt{4x^2 + 9}},\ 0 \leqslant x \leqslant 5$$

Given that 1 unit on each axis represents 10 metres, use integration to calculate the area of R. Show your method clearly and give your answer to 2 significant figures. **(6 marks)**

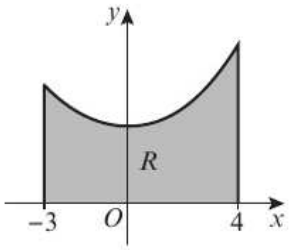
E/P **21** Find $\displaystyle\int \frac{1}{\sqrt{x^2 - 2x + 10}}\,\mathrm{d}x$ **(7 marks)**

E/P **22** **a** Find $\displaystyle\int \frac{1}{\sinh x + 2\cosh x}\,\mathrm{d}x$ **(6 marks)**

b Show that $\displaystyle\int_1^4 \frac{3x - 1}{\sqrt{x^2 - 2x + 10}}\,\mathrm{d}x = 9(\sqrt{2} - 1) + 2\,\mathrm{arsinh}\,1$ **(6 marks)**

E/P **23** The diagram shows a hanging chain fixed to two vertical bollards fixed on horizontal ground 7 metres apart.

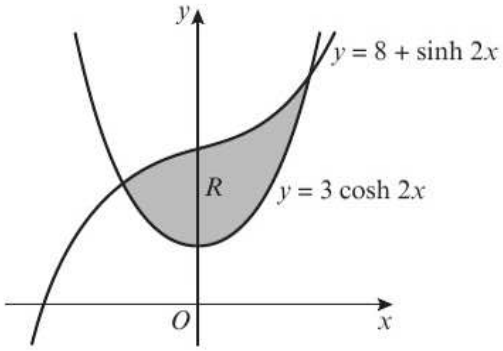
The chain is modelled by the curve $y = \cosh\frac{x}{3}$



a Find the acute angle that the chain makes with the right-hand bollard. Give your answer to 3 significant figures. **(5 marks)**

b Use calculus to find, correct to 2 decimal places, the area of the finite region, R, enclosed by the bollards, the chain and the ground. **(5 marks)**

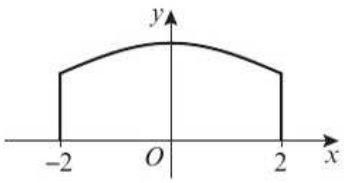
E **24** The curves shown have equations $y = 3\cosh 2x$ and $y = 8 + \sinh 2x$.



a Using the definitions of $\sinh x$ and $\cosh x$ in terms of e^x, find exact values for the x-coordinates of the two points where the curves intersect. **(6 marks)**

b Use calculus to find the area of the finite region, R, between the two curves, giving your answer correct to 3 significant figures. **(6 marks)**

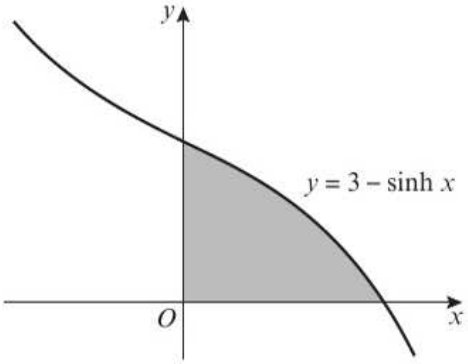
E/P **25** The diagram shows the cross-section of a loaf of bread. Each unit on the axes represents 5 cm. The curved top of the loaf is modelled using the equation



$$y = \frac{5}{\sqrt{x^2 + 4}}$$

Given that the loaf can be modelled as a prism, with length 30 cm, find, correct to 3 significant figures, the volume of the loaf. **(8 marks)**

E **26** The diagram shows the curve $y = 3 - \sinh x$.



a Find the exact coordinates of the point where the curve crosses the x-axis. **(3 marks)**

The shaded area is bounded by the curve, the y-axis and the x-axis.

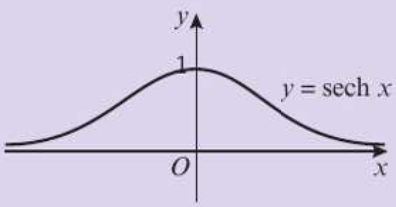
b Find the volume of the solid of revolution formed when the shaded area is rotated through 360° about the x-axis. Give your answer correct to 3 significant figures. **(8 marks)**

Challenge

The diagram shows the graph of

$y = \text{sech}\, x$.

Show that the area bounded by the curve and the x-axis is π.



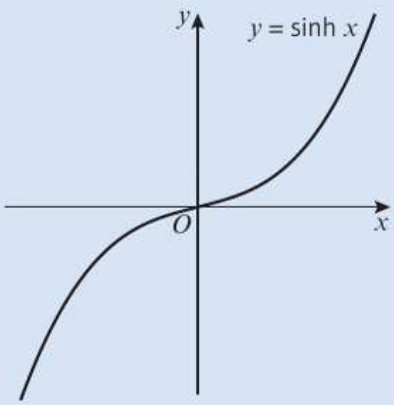
Summary of key points

1 • Hyperbolic sine (or **sinh**) is defined as $\sinh x \equiv \frac{e^x - e^{-x}}{2}, x \in \mathbb{R}$

• Hyperbolic cosine (or **cosh**) is defined as $\cosh x \equiv \frac{e^x + e^{-x}}{2}, x \in \mathbb{R}$

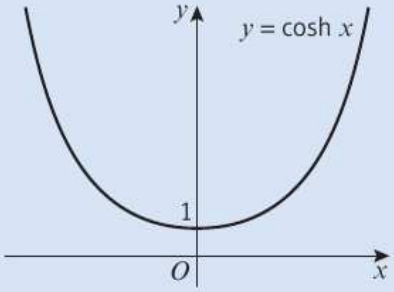
• Hyperbolic tangent (or **tanh**) is defined as $\tanh x \equiv \frac{\sinh x}{\cosh x} \equiv \frac{e^{2x} - 1}{e^{2x} + 1}, x \in \mathbb{R}$

2 • The graph of $y = \sinh x$:



For any value a, $\sinh(-a) = -\sinh a$.

• The graph of $y = \cosh x$:



For any value a, $\cosh(-a) = \cosh a$.

3 The table shows the inverse hyperbolic functions, with domains restricted where necessary.

Hyperbolic function	Inverse hyperbolic function
$y = \sinh x$	$y = \text{arsinh}\, x$
$y = \cosh x, x \geqslant 0$	$y = \text{arcosh}\, x, x \geqslant 1$
$y = \tanh x$	$y = \text{artanh}\, x, \lvert x\rvert < 1$

4 • $\text{arsinh}\, x = \ln\left(x + \sqrt{x^2 + 1}\right)$ • $\text{arcosh}\, x = \ln\left(x + \sqrt{x^2 - 1}\right), x \geqslant 1$ • $\text{artanh}\, x = \frac{1}{2}\ln\left(\frac{1 + x}{1 - x}\right), \lvert x\rvert < 1$

5 $\cosh^2 A - \sinh^2 A \equiv 1$

6 • $\sinh(A \pm B) \equiv \sinh A \cosh B \pm \cosh A \sinh B$

• $\cosh(A \pm B) \equiv \cosh A \cosh B \pm \sinh A \sinh B$

7 • $\frac{d}{dx}(\sinh x) = \cosh x$ • $\frac{d}{dx}(\cosh x) = \sinh x$ • $\frac{d}{dx}(\tanh x) = \text{sech}^2 x$

• $\frac{d}{dx}(\text{arsinh}\, x) = \frac{1}{\sqrt{x^2 + 1}}$ • $\frac{d}{dx}(\text{arcosh}\, x) = \frac{1}{\sqrt{x^2 - 1}}$ • $\frac{d}{dx}(\text{artanh}\, x) = \frac{1}{1 - x^2}$

8 • $\int \sinh x\, dx = \cosh x + c$ • $\int \cosh x\, dx = \sinh x + c$

9 $\int \tanh x\, dx = \ln \cosh x + c$

10 • $\int \frac{1}{\sqrt{a^2 + x^2}} dx = \text{arsinh}\left(\frac{x}{a}\right) + c$ • $\int \frac{1}{\sqrt{x^2 - a^2}} dx = \text{arcosh}\left(\frac{x}{a}\right) + c, \ x > a$